Testicular germ cell tumor

Most common solid malignancy in males 15–35 · >90% cure, >95% 5-yr survival — chemotherapy is curative even in advanced disease · lung is the commonest met site (62%); liver (16%) and brain (7%) are negative prognostic factors

Stanford Oncology Handbook 2025, Testicular Cancer (Moore K, Glover M, Shah S) · NCCN v1.2025 · trial data verified against the primary publications

WORKUP

- Suspicious mass: H&P, AFP, bHCG, LDH, chemistry, testicular ultrasound
- **Repeat all markers after orchiectomy** — TNM staging uses post-orchiectomy values
- **Pure seminoma** (seminoma histology + normal AFP; bHCG may be elevated): abdominal/pelvic CT + chest X-ray; chest CT only if abdominal CT positive or CXR abnormal
- **Non-seminoma** (incl. mixed, and seminoma histology with elevated AFP): chest/abdominal/pelvic CT up front
- Both: brain MRI if clinically indicated · **sperm banking**
- Nodal spread is predictable — right-sided to inter-aortocaval, left-sided to para-aortic

HISTOLOGY → MARKER PATTERN

Subtype	Markers
Seminoma	AFP **always normal**; hCG ↑ in ~15–20%
Embryonal carcinoma	AFP and/or hCG; commonest mixed component
Yolk sac tumor	**AFP** — the classic AFP producer
Choriocarcinoma	hCG, often >50,000; hemorrhage, brain mets
Teratoma	**neither** — marker-negative, chemoresistant
Mixed germ cell tumor	reflects whichever components are present

Seminoma hCG comes from syncytiotrophoblasts and is compatible with pure seminoma — **only AFP reclassifies**. Very high hCG in choriocarcinoma can cause hyperthyroidism (TSH receptor cross-reactivity) and gynecomastia. LDH is non-specific and tracks tumor bulk. Teratoma is marker-negative and chemoresistant — the reason post-chemotherapy masses must be resected.

SERUM TUMOR MARKERS (S)

S	LDH	hCG	AFP
S1	<1.5× N	<5,000	<1,000
S2	1.5–10× N	5,000–50,000	1,000–10,000
S3	>10× N	>50,000	>10,000

hCG mIU/mL, AFP ng/mL. pT1a/pT1b (<3 vs ≥3 cm) applies to pure seminoma only. cN1 ≤2 cm · cN2 >2–5 cm · cN3 >5 cm. M1a nodal/pulmonary · M1b non-pulmonary visceral.

IGCCCG RISK

Non-seminoma	**Good:** testis/retroperitoneal primary, no non-pulmonary visceral mets, AFP<1,000 + hCG<5,000 + LDH<1.5×. **Intermediate:** markers in the middle band. **Poor:** mediastinal primary, non-pulmonary visceral mets, or AFP>10,000 / hCG>50,000 / LDH>10×
Seminoma	**Good:** any primary, no non-pulmonary visceral mets, normal AFP. **Intermediate:** non-pulmonary visceral mets. **There is no poor-risk category** — only AFP matters, not hCG or LDH

TREATMENT PATHWAY

Stage I

Seminoma
~85% cured by orchiectomy alone
surveillance strongly preferred
alt: carboplatin ×1–2 or RT (risk factor >4 cm)

Non-seminoma
orchiectomy cures ~70%; relapses highly curable
surveillance if no risk factors
RFs (LVI, cord/scrotal, embryonal): **BEP ×1**

Stage IIA–IIB

Both histologies
BEP ×3 or EP ×4
seminoma: RT or RPLND if IIA / non-bulky IIB
non-seminoma: or nerve-sparing RPLND

Stage IIC–III

By IGCCCG risk
good risk: **BEP ×3 or EP ×4**
intermediate or poor: **BEP ×4 or VIP ×4**
seminoma IIC treated as BEP ×3 / EP ×4

Residual mass and relapse

Rules differ by histology
seminoma >3 cm: **PET** — avid resect, non-avid observe
non-seminoma: **resect ALL masses >1 cm**
~10% viable cancer, 40% teratoma, 50% necrosis
relapse: resect, else TIP/VeIP ×4 or HDCT

PEARLS

- **The AFP rule overrides histology.** Pure seminoma requires seminoma histology **and** a normal AFP; seminoma histology with an elevated AFP is worked up and treated as a non-seminoma. Elevated bHCG is compatible with pure seminoma
- **Dose holds and reductions are discouraged in the curative (1st and 2nd line) setting** — delay only for severe active infection or illness. The one stated exception: hold or reduce cisplatin for creatinine >1.5 (EP)
- **Residual mass rules differ by histology** — PET works for seminoma but not non-seminoma, because teratoma is not FDG-avid yet is chemoresistant and must be resected
- Non-pulmonary visceral metastases define **intermediate-risk seminoma** and **poor-risk non-seminoma**
- RT or RPLND remains an option only for stage IIA and non-bulky IIB seminoma (nodes <3 cm)
- Sperm banking before treatment; restage markers after orchiectomy

REGIMENS & DOSING

Regimen	Dosing	Cycle
BEP (Einhorn)	cisplatin 20/day d1–5 · etoposide 100/day d1–5 · bleomycin 30 units d1, 8, 15 · filgrastim d6–12	q21d
EP	etoposide 100/day d1–5 · cisplatin 20/day d1–5 · pegfilgrastim d5→6	q21d
VIP	etoposide 75 · ifosfamide 1200 · cisplatin 20, all d1–5 · mesna · pegfilgrastim d6	q21d
TIP (2L)	paclitaxel 250 d1 · ifosfamide 1500 d2–5 · cisplatin 25 d2–5 · mesna	q21d
VeIP (2L)	vinblastine 0.11 mg/kg d1–2 · ifosfamide 1200 d1–5 · cisplatin 20 d1–5 · mesna	q21d
HDCT	carboplatin 700 + etoposide 750 — days –5, –4, –3 before stem-cell infusion, ×2 cycles	with PBSC rescue
3L options	gemcitabine + paclitaxel ± oxaliplatin · oxaliplatin + irinotecan · oral etoposide 50–100 mg d1–21	
Pembrolizumab	200 mg q3wk (or 400 mg q6wk) — MSI-H, dMMR, or TMB >10	q3wk

All doses mg/m² unless stated. Growth factor support is built into every first- and second-line regimen; mesna with every ifosfamide-containing regimen.

LANDMARK TRIALS

Trial	Result
IGCCCG 1997 · upd. 2021	2021 (n=9,728): 5-yr OS good **96%** / int 89% / poor **67%** (was 91/79/48); adds LDH >2.5× ULN
Einhorn 1989 (SECSG)	**3 vs 4 BEP**, favorable risk — NED 98% vs 97%. n=184
EORTC 30941 / TE20	**3 vs 4 BEP** and 5- vs 3-day — 2-yr PFS 90.4% vs 89.4%, equivalent
Kondagunta 4×EP	Single-arm n=289, good risk — CR **98%**. Bleomycin-free option
MRC TE19 / EORTC 30982	Stage I seminoma: **carboplatin AUC7 ×1 non-inferior to RT** (5-yr RFS 94.7% vs 96.0%)
MRC TE18 / EORTC 30942	Seminoma RT **20 Gy = 30 Gy**, less lethargy
SWENOTECA	CS I non-sem **1×BEP**: 5-yr relapse **3.2% LVI+** / **1.6% LVI–** vs 41.7% / 13.2% surveillance
Kondagunta TIP	First salvage n=46 — durable CR **63%**
TI-CE (Feldman)	High-dose + PBSC, n=107 — CR 50%, **5-yr DFS 47%**
TIGER (A031102)	Ongoing phase 3, **TIP vs TI-CE** at first salvage, primary OS

Phase 3 only for the BEP-cycle and seminoma-RT questions. SWENOTECA is population-based; Kondagunta TIP (ph 2), 4×EP and TI-CE (ph 1/2) are single-arm.

> **Biomarkers:** testicular cancer is **marker-driven, not biomarker-driven** — AFP, bHCG and LDH determine stage, risk group and response. The only biomarker-selected therapy in the chapter is pembrolizumab for MSI-H, dMMR, or TMB >10.